The Fair Pricing Playbook

A practical framework for developing, evaluating, and auditing fair algorithmic pricing systems

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TECHNICAL REPORT

A practical framework for fair algorithmic pricing in insurance and financial services

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1. Abstract

Algorithmic pricing is now common in insurance and financial services, but practical guidance for translating fairness principles into deployable workflows remains fragmented. This preprint presents The Fair Pricing Playbook, a four-step framework linking fairness definition, model design, welfare assessment, and post-deployment audit into one end-to-end process. The framework centers the anti-discrimination lens while keeping actuarial fairness, solidarity, and consumer trust visible in governance decisions.

Three technical case studies provide implementation evidence. Case Study 1 applies five anti-discrimination model designs — M0, MU, MDP, MCDP, and MC — to French motor insurance data using GLMs and XGBoost, comparing fairness and predictive accuracy across criteria. Case Study 2 extends the analysis to the full pricing pipeline, evaluating five pricing regulations (P0, PA, POB, PDP, PAF) in terms of consumer welfare and firm profit by gender on a 2,000-policyholder French subsample. Case Study 3 implements conditional demographic parity (CDP) and proxy discrimination (PD) audits with corrected statistical inference on Illinois auto insurance quote data, demonstrating that classical OLS standard errors can be substantially wrong for deterministic pricing outputs.

The goal is operational clarity: how to choose a fairness criterion, map it to modeling interventions, evaluate market consequences, and run statistically valid compliance audits. The full playbook, including interactive case studies, implementation checklists, and reproducible code, is available at fair.feihuang.org.

Keywords: algorithmic pricing, insurance, fairness, anti-discrimination, welfare analysis, audit, responsible AI.

2. Executive Summary

AI-driven pricing systems are now standard in insurance and financial services. They improve risk segmentation and speed decisions, but can also reproduce or amplify discriminatory patterns through proxy variables, complex models, and opaque optimization pipelines. Regulators in the United States (Colorado, New York), the European Union, and Australia are responding with new requirements for fairness testing and algorithmic accountability. Practitioners across compliance, actuarial, data science, and governance functions need a coherent workflow — not isolated tools.

This report presents The Fair Pricing Playbook: a four-step framework that links fairness definition, model design, welfare assessment, and post-deployment audit into one end-to-end process, supported by three reproducible case studies in insurance.

Table 1: The four steps of the Fair Pricing Playbook.

Step	Question	Key output
1 — Define fairness	What does fair mean here?	Documented criterion, scope, and legitimate factors

Step	Question	Key output
2 — Design fair pricing	How can the criterion be achieved?	Fair cost model (MU / MDP / MCDP / MC)
3 — Assess impact	Who gains and loses once prices are set?	Welfare and profit analysis by protected group
4 — Audit the system	Does the deployed system pass or fail?	Pre-committed verdict: Pass / Fail / Insufficient information

Why each step is necessary. The steps are sequential and cannot be reordered or skipped. The criterion chosen in Step 1 determines which models are permissible in Step 2. Cost-model fairness does not guarantee premium fairness or consumer welfare improvements once demand, competition, and regulation are taken into account (Step 3). And audit conclusions in Step 4 depend critically on whether statistical inference is correctly specified for deterministic algorithmic outputs — a point that standard practice currently ignores.

Key findings from the three case studies:

- **Cost-model fairness premium fairness.** Applying fairness interventions at the cost-modeling stage can alter final premiums in ways that reduce welfare for the protected group, through selection effects and demand response. Evaluation must extend to the full pricing pipeline.

- **Premium fairness and markup fairness are in tension.** No single pricing regulation simultaneously achieves equal premiums and equal markups across protected groups. PDP (demographic parity pricing) closes premium gaps but increases female markups in the French motor insurance data. PA (accountable pricing) reduces markup gaps but generates larger profit losses. Regulators and firms should specify which metric they are targeting and evaluate proposals on all dimensions.

- **Classical audit inference is unreliable for algorithmic pricing.** Deterministic pricing algorithms produce residuals that represent approximation error, not sampling variability. Classical OLS standard errors are wrong in both magnitude and direction — in the Illinois auto insurance data, HC3-to-classical ratios range below 0.5 and above 1.5 across companies. The direction cannot be predicted in advance, and the audit verdict can change depending on which standard error is used.

- **Insufficient information is not a pass.** Under equivalence testing (TOST), compliance is established only when statistical evidence shows the price gap falls within pre-specified tolerance bands. A non-significant conventional test only means no gap was detected — it does not demonstrate fairness.

Who this is for. Compliance leaders and regulators: Executive Summary → Steps 1 and 4 → Section 7. Actuaries and pricing professionals: Steps 1–3 → Case Studies 1 and 2. Data scientists: Steps 2 and 4 → Case Studies 1 and 3. Policy and government: Executive Summary → Step 3 → Section 7.

The full playbook — interactive guidance, implementation checklists, and reproducible code in R and Python — is freely available at **fair.feihuang.org**.

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3. 1. Introduction

AI-driven pricing systems are now widely used in insurance and other financial services. Their appeal is clear: better risk segmentation, faster decisions, and potentially better matching between risk and price. At the same time, these systems can reproduce or amplify discriminatory patterns through proxy variables, complex model interactions, and opaque optimization pipelines. This creates a governance gap between predictive performance and fairness accountability.

In regulated settings, fairness is not a single model constraint. It is a sequence of decisions spanning criterion selection, model design, market behavior, and post-deployment testing (Frees and Huang 2023; Xin and Huang 2024). A model can satisfy one fairness notion and still produce adverse outcomes once final prices incorporate loadings, demand response, and regulatory rules (Huang, Shimao, and Khern-am-nuai 2026; Huang and Shimao 2026). Likewise, an audit can fail if statistical inference is mis-specified even when design intent is sound (Huang and Hooker 2026).

Regulatory attention has intensified across jurisdictions. In the United States, Colorado's Division of Insurance and New York's Department of Financial Services have each issued proposals requiring insurers to evaluate whether algorithmic pricing systems produce discriminatory pricing outcomes (Colorado Division of Insurance 2023; New York State Department of Financial Services 2024). These proposals operationalize fairness tests — conditional demographic parity and proxy discrimination — that map directly to the audit methodology in Step 4 of this framework.

This preprint introduces The Fair Pricing Playbook, a practical four-step framework linking these decisions into one auditable workflow. Section 2 situates the work in related literature. Section 3 gives the framework overview. Section 4 develops the methodological core of each step in depth, including formal definitions, model taxonomies, welfare measures, and audit statistics. Section 5 describes the companion case studies in detail, including data, implementation choices, and illustrative findings. Section 6 summarizes contributions, and Section 7 discusses limitations.

4. 2. Background and Related Literature

4.1 2.1 Fairness in Insurance Pricing

What makes a price fair? Frees and Huang (2023) frame the question as: what are the principles for assessing the appropriateness of insurance differentiation? They distinguish risk-based differentiation — premiums that differ because expected losses differ — from non-risk differentiation, where prices diverge for the same underlying risk due to retention modeling, willingness to pay, or price optimization. Risk-based rating is typically defended on efficiency grounds; non-risk differentiation raises fairness concerns when it reproduces group disparities or penalizes loyalty.

Xin and Huang (2024) formalize anti-discrimination principles for insurance pricing, distinguishing direct discrimination (explicit use of protected attributes) from indirect discrimination (disparate outcomes through proxy variables or complex model interactions). They propose a taxonomy of model designs — M0, MU, MDP, MCDP, MC — that link specific fairness criteria to concrete modeling interventions, and show that omitting protected attributes (fairness through unawareness) does not prevent indirect discrimination if correlated proxies remain.

Krafcheck, Balnozan, and Huang (2026) extend this taxonomy to life insurance, surveying a broader set of individual and group fairness criteria and mapping them to the independence, separation, and sufficiency frameworks from the algorithmic fairness literature. A key finding is that individual and group fairness criteria are often mutually incompatible: satisfying one typically requires violating another.

4.2 2.2 Welfare and Market Effects of Fairness Interventions

A separate literature examines what happens to consumers and insurers when fairness constraints are applied. Huang, Shimao, and Khern-am-nuai (2026) show that fairness interventions at the cost-model stage do not translate uniformly into welfare improvements at the market stage. Using a discrete-choice demand framework applied to French motor insurance data, they find that some standard fairness metrics can *reduce* welfare for the protected group after selection effects and demand response are taken into account, and that firm vs consumer trade-offs vary substantially by intervention type.

Huang and Shimao (2026) develop a complete pricing pipeline model that incorporates cost modeling, demand modeling, price optimization, and regulatory constraints, and evaluate five pricing regulations (P0, PA, POB, PDP, PAF) in terms of premium gaps, markup gaps, consumer welfare, and insurer profit. A central finding is that premium fairness and markup fairness are in tension: no single regulation achieves both simultaneously. PDP closes premium gaps but can widen markup disparities for the group it is designed to protect.

4.3 2.3 Audit Methodology for Algorithmic Pricing

Huang and Hooker (2026) develop the audit methodology used in Step 4. They identify a key statistical problem: deterministic pricing algorithms always return the same price for the same input profile, so audit regression residuals represent model approximation error rather than independent sampling variability. Classical OLS standard errors — which assume i.i.d. errors — can be wrong in both magnitude and direction. They prescribe HC3 sandwich standard errors for CDP audits and a full cross-covariance correction for proxy discrimination tests. They also argue for equivalence testing (TOST) rather than conventional significance testing for compliance auditing, because a non-significant result under conventional testing is not positive evidence of fairness — it only means a gap was not detected.

Xin, Hooker, and Huang (2026) examine the consequences of using proxied protected attributes (BISG, BIFSG, zip-code composition) in audit regressions. Proxy misclassification mixes group effects, and estimated gaps can be attenuated toward the majority group even when overall proxy accuracy appears high. Proxy errors can correlate with geography and pricing residuals, further distorting disparity estimates.

4.4 2.4 Positioning

Compared with these contributions, the playbook's role is integrative. It provides an operational bridge from policy framing to modeling choices, welfare analysis, and audit execution, designed to be used by practitioners across compliance, actuarial, data science, and governance functions. It does not derive new statistical theory but assembles the decisions that must be made in sequence — and in a coherent order — to produce a defensible fair-pricing system.

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5. 3. Framework Overview

The framework has four linked steps. Table 1 summarizes each step, the governing question it answers, and the type of output it produces.

Table 2: Table 1. The four steps of the Fair Pricing Playbook.

Step	Topic	Governing question	Primary output
1	Define fairness	What does fair mean here?	Documented criterion, scope, legitimate factors
2	Design fair pricing	How can the criterion be achieved?	Fair cost model (M0/MU/MDP/MCDP/MC)
3	Assess impact	Who gains and loses after prices are set?	Welfare and profit analysis by protected group
4	Audit the system	Does the deployed system pass or fail?	Pre-committed audit verdict: pass / fail / insufficient information

The logic is sequential, with each step providing inputs to the next. Step 1 fixes the fairness criterion and governance scope before any modeling begins — this choice constrains what model designs are permissible in Step 2 and what tests are run in Step 4. Step 2 translates that criterion into a model intervention. Step 3 evaluates whether cost-model fairness carries through to final prices. Step 4 applies pre-committed statistical tests to deployed pricing outputs.

Three design principles underpin the framework. First, protocol choices must be fixed before reviewing outcomes: post-hoc changes to criteria, tolerance margins, or sample definitions undermine auditability and stated error-rate control. Second, fairness evaluation must span the full pricing pipeline, not stop at the cost model — welfare analysis in Step 3 is not optional. Third, audit inference must match the statistical properties of the outputs: deterministic algorithmic pricing requires corrected standard errors.

Figure 1 illustrates the pathway from model design through fairness testing, as displayed on the companion website.

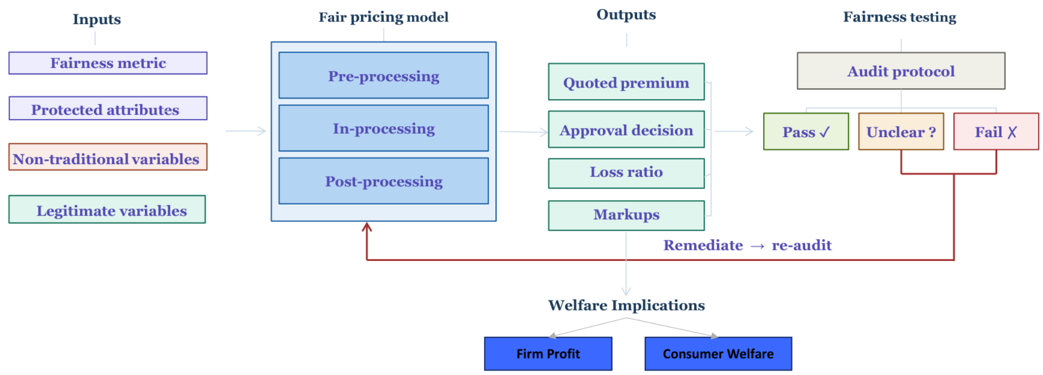

Figure 1: Pathway towards fairness: from model inputs and design through outputs, welfare, and fairness testing to pass, unclear, or fail outcomes. Source: fair.feihuang.org

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6. 4. Methodological Core

6.1 4.1 Step 1: Define Fairness Criteria and Governance Scope

6.1.1 4.1.1 Three lenses

Step 1 organizes the fairness discussion around three lenses. The *anti-discrimination* lens asks whether pricing complies with anti-discrimination law on how people are rated and what outcomes groups receive — this is the primary technical focus of Steps 2–4. The *competing ethical notions* lens asks which differentiations are appropriate given how a product is positioned: insurance as a social good (solidarity, community rating) vs an economic commodity (risk-based, adverse-selection-limiting). The *consumer trust* lens asks whether pricing honors what the firm has promised through product disclosure, marketing, and conduct — compliance with anti-discrimination rules is necessary but not sufficient for this lens.

6.1.2 4.1.2 Discrimination types

Following Xin and Huang (2024), *direct discrimination* occurs when a person is treated less favorably because a protected characteristic differs; if the protected attribute is excluded from the rating model, direct discrimination can be prevented. *Indirect discrimination* arises after direct discrimination is ruled out: a person may still receive disproportionate outcomes because protected status is inferred through apparently neutral variables. Two channels matter in algorithmic pricing:

- **Identifiable proxies**: variables correlated with a protected attribute, such as geographic area correlated with race, or variables correlated with gender.

- **Unidentifiable proxies**: opaque models that combine many variables and reproduce protected-group disparities without a single obvious surrogate.

Algorithmic discrimination is generally a form of indirect discrimination. Omitting protected variables (fairness through unawareness) does not guarantee fair outputs if correlated proxies or complex nonlinear models remain.

6.1.3 4.1.3 Formal fairness criteria

For readability, this report uses plain language where possible. Protected attribute means the protected class variable, non-protected features are other rating inputs, legitimate factors are regulator-accepted risk controls, and predicted outcome means the model's predicted cost or price (Xin and Huang 2024).

The **playbook criteria** — those implemented in Steps 2–4 — are:

Table 3: Table 2. Fairness criteria implemented in the playbook.

Type	Criterion	Definition
Individual	Fairness through unawareness (FTU)	The model does not use the protected attribute directly
Individual	Controlling for the protected variable (CPV)	Predictions are averaged across protected-group membership at scoring time
Group	Demographic parity (DP)	Predicted outcomes are equal on average across protected groups
Group	Conditional demographic parity (CDP)	Predicted outcomes are equal across protected groups after controlling for legitimate factors
Group	Separation / sufficiency	Equal prediction errors across groups (separation) or equal calibration (sufficiency)

A broader taxonomy covering individual fairness (fairness through awareness, no omitted-variable bias) and group fairness (independence, separation, sufficiency) is surveyed in Krafcheck, Balnozan, and Huang (2026) for life insurance settings.

Achieving individual and group fairness simultaneously is often impossible (Krafcheck, Balnozan, and Huang 2026; Xin and Huang 2024). The choice of criterion must precede modeling. Prohibition on protected variables typically points toward individual criteria (FTU, CPV); disparate-impact rules point toward group criteria (DP, CDP). Colorado's and New York's regulatory proposals both operationalize CDP-adjacent and proxy-discrimination tests, making CDP and PD the natural illustration criteria for this framework.

6.1.4 4.1.4 Legitimate factors

Legitimate factors are risk-related variables the applicable regulator accepts as grounds for price variation. They carry forward unchanged into the MCDP model (Step 2) and serve as controls in the Step 4 audit. Their identification and legal/compliance sign-off are required outputs of Step 1.

6.2 4.2 Step 2: Translate Criteria into Model Interventions

6.2.1 4.2.1 Model taxonomy

Step 2 translates the criterion from Step 1 into a model intervention at the cost-modeling stage (pure premium or expected claim cost). Following Xin and Huang (2024), five standard model designs are defined:

Table 4: Table 3. Model taxonomy linking fairness criteria to interventions.

Model	Criterion	Intervention stage	Approach
M0 (full)	Baseline	—	Uses all predictors including protected attributes. Kept for comparison, not deployment where prohibited
MU (un-awareness)	FTU	—	Uses non-protected predictors only; protected attribute excluded
MDP (demographic parity)	DP	Pre-processing	Debiases all non-protected predictors before fitting

Model	Criterion	Intervention stage	Approach
MCDP (conditional demographic parity)	CDP	Pre-processing	Keeps legitimate predictors unchanged and debiases non-legitimate predictors
MC (controlling for protected variable)	CPV	Post-processing	Fits full model, then averages predictions over protected-group membership at scoring

M0 is retained for welfare and audit comparisons in Steps 3 and 4. MU is the common industry baseline (excluding the protected attribute but making no further intervention). The distinction between MDP and MCDP is substantive: MDP debiases all non-protected predictors, targeting unconditional demographic parity; MCDP retains legitimate predictors unchanged and debiases only non-legitimate predictors, targeting conditional demographic parity that allows group differences only through legitimate factors.

6.2.2 4.2.2 Pre-processing: debiased predictors

MDP and MCDP use disparate-impact removal (Feldman et al. 2015) to construct debiased predictors before fitting. The procedure aligns predictor distributions across protected groups while preserving within-group rank ordering. For ordinal variables (for example bonus group and age), small jitter is added before debiasing and the result is mapped back to factor levels.

For MCDP, debiasing is applied only to non-legitimate predictors. Legitimate predictors enter unchanged, allowing group differences that derive from regulator-accepted risk factors. This maps to a conditional demographic parity target: after controlling for legitimate factors, predicted costs should be approximately equal across protected groups.

6.2.3 4.2.3 Post-processing: CPV averaging

MC achieves CPV by fitting a full model and then averaging predictions over protected-group membership at scoring time. This keeps the predictive information in non-protected features while removing the direct marginal effect of protected-group assignment in the final score. Xin and Huang

(2024) show that MC often controls proxy effects better than MU because it explicitly accounts for correlation between protected and non-protected predictors.

6.2.4 4.2.4 In-processing

In-processing adds fairness constraints directly into the training or optimization objective. This approach is illustrated in Case Study 2 for price optimization (PDP and PAF pricing rules), where demographic parity constraints are enforced via penalty terms. At the cost-modeling stage, in-processing can take the form of regularization or Lagrangian constraints that penalize group-level prediction disparities during training.

6.2.5 4.2.5 Regulations, criteria, and models

Xin and Huang (2024) map regulatory regimes to fairness criteria and model designs. Input-based regulation (prohibiting specific variables) typically points toward FTU or CPV → MU or MC. Effects-based or outcome-based regulation (quantitative tests on premiums or loss ratios) points toward DP or CDP → MDP or MCDP. The criterion that is legally binding should be determined in Step 1 before selecting the model in Step 2.

6.3 4.3 Step 3: Evaluate Welfare Implications at the Final-Price Stage

6.3.1 4.3.1 Why cost-model fairness is insufficient

A fair cost model does not guarantee a fair or welfare-improving final price. Huang, Shimao, and Khern-am-nuai (2026) demonstrate that fairness interventions at the cost-modeling stage can alter final prices in ways that reduce welfare for the protected group, even when the intervention appears to reduce premium disparities at the cost-model level. The mechanisms include: (a) selection effects — when price changes affect who purchases insurance, the composition of the insured pool changes in ways that can harm the group the intervention was designed to help; (b) markup effects — because firms optimize over consumer demand, equalizing estimated costs does not equalize markups; and (c) demand-elasticity heterogeneity — groups that are more price-sensitive may be disproportionately affected by pricing rule changes.

6.3.2 4.3.2 The full pricing pipeline

Huang and Shimao (2026) model the complete pipeline from cost modeling through market outcomes:

Stage	Role
Cost modeling	Expected claim cost (input to pricing)
Demand modeling	Purchase probability as a function of offered premium
Price optimization	Offered premium p_i that maximizes expected profit subject to regulatory constraint
Regulatory constraints	Fairness and accountability rules that restrict the pricing rule

Consumer welfare is measured using certainty equivalence, the sure dollar amount making a consumer indifferent to buying or not buying insurance. Firm profit is the sum of purchased premiums minus expected claim cost. Markup is premium relative to expected cost.

6.3.3 4.3.3 Pricing rules

Five pricing rules are evaluated, representing a spectrum from unconstrained to fairness-constrained:

Table 6: Table 4. Pricing rules and their regulatory interpretations.

Rule	Regulatory interpretation	Constraint
P0	Unconstrained profit maximization	None — benchmark
PA	Accountable pricing	Premiums follow a transparent, decomposable rating structure
POB	Price optimization ban	Premium is tied linearly to estimated cost
PDP	Demographic parity on premiums	Average premiums are equal across protected groups
PAF	Actuarial group fairness	Premiums are equal across protected groups within each cost stratum

PDP and PAF represent fairness constraints at the premium level: PDP targets unconditional equality of mean premiums across groups; PAF targets equality within cost strata, allowing group differences only through estimated risk. POB and PA represent transparency or accountability constraints rather than explicit fairness requirements.

6.3.4 4.3.4 Key findings from the welfare literature

Huang and Shimao (2026) find that premium fairness and markup fairness are in fundamental tension. Under unconstrained pricing (P0), female policyholders in the French data face lower premiums on average (because they have lower estimated costs) but higher markups (because the insurer extracts more profit from them relative to their risk). PDP moves closest to premium equality but increases markup disparity. PA reduces markup gaps but generates larger profit losses. POB effects depend on market structure (mandatory vs. voluntary participation, degree of competition). No single regulation achieves both premium equality and markup equality simultaneously.

These findings have direct governance implications: firms and regulators should specify *which* metric — premium, markup, consumer welfare, or approval rate — they are targeting, and should evaluate proposed regulations on all relevant dimensions before implementation.

6.4 4.4 Step 4: Audit with Corrected Inference

6.4.1 4.4.1 The pre-committed audit protocol

Step 4 applies a Plan → Audit → Decide → Improve protocol (Huang and Hooker 2026). All design choices must be fixed before examining pricing data:

1. **Select the fairness criterion**: CDP for system-level price gaps after legitimate controls; PD when screening a suspect rating variable; other criteria from Step 1 if required by law.
2. **Specify the protected attribute** - directly observed or responsibly proxied, with method and limitations documented.
3. **Specify legitimate rating factors** aligned with Step 1 and regulatory guidance.
4. **Define the response variable** - quoted premium, pure premium, or loss ratio. This determines the meaning of any measured disparity.
5. **Set tolerance margins** for CDP, such as a level-gap margin (for example 5% of mean premium), ratio margin (for example 0.80 adverse-impact ratio), and significance level.
6. **Specify the standard error estimator**: HC3 for CDP; full cross-covariance for PD.
7. **Design a representative quote sample**: stratified profiles submitted within a short time window, with sample size planned for adequate power.

Post-hoc changes to any of these choices undermine stated error-rate control and auditability.

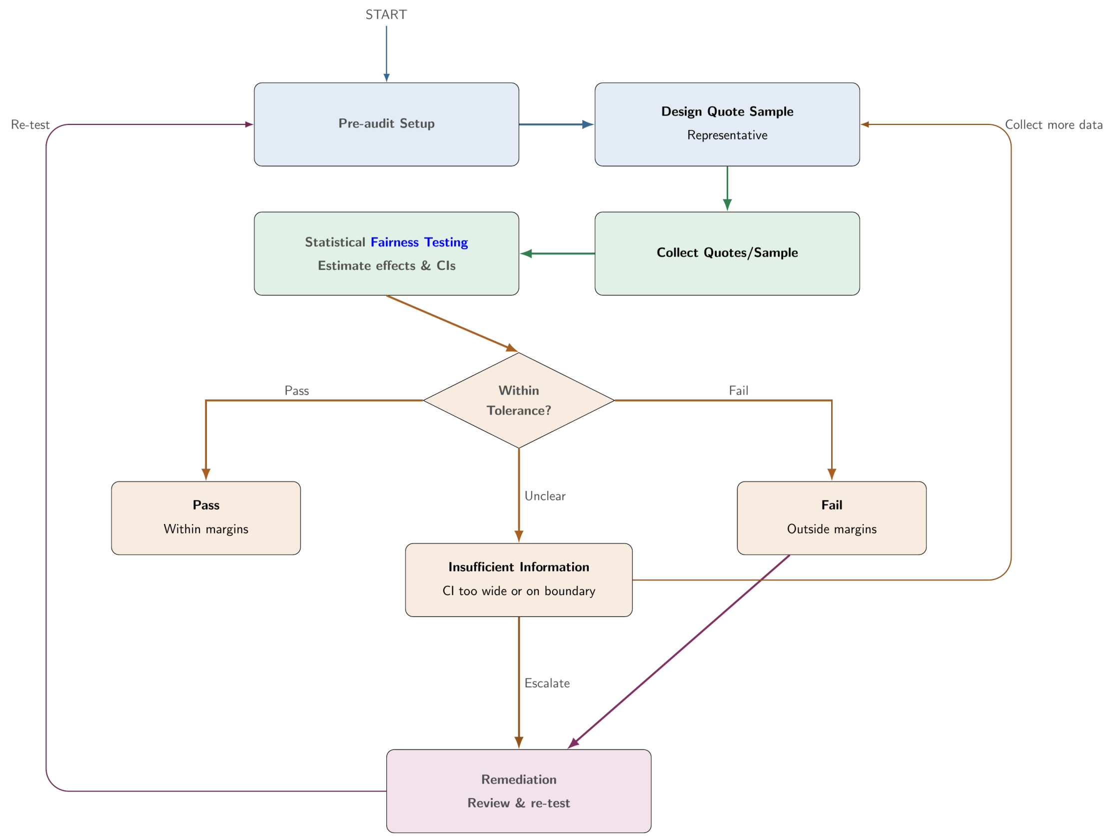

Figure 2: End-to-end fairness audit flow: Plan, Audit, Decide, Improve. Source: fair.feihuang.org

6.4.2 4.4.2 Why classical OLS standard errors fail for algorithmic pricing

The CDP audit regresses quoted premiums on protected-group status and legitimate risk controls. The key quantity is the conditional premium gap after those controls.

For deterministic pricing algorithms, the same input profile always returns the same price. Residuals therefore reflect model approximation error rather than independent sampling noise. Classical OLS standard errors assume independent, identically distributed random errors, which is not appropriate in this setting.

The practical fix is to use HC3 robust standard errors (MacKinnon and White 1985), which are built from observation-level residual behavior and account for leverage. For proxy-discrimination tests, a second correction is needed: because both models are fitted to the same premium vector, their estimates are correlated. Ignoring this cross-covariance overstates uncertainty in the coefficient shift and can make the proxy test overly conservative.

As shown in Case Study 3 below, the ratio of HC3 to classical standard errors varies substantially across companies (ranging below 0.5 and above 1.5 in the Illinois data), and the direction cannot be predicted in advance. The audit verdict can change depending on which standard error is used.

6.4.3 4.4.3 Equivalence testing (TOST) for CDP

Conventional significance testing asks whether a price gap is detectably different from zero. This is a poor fit for compliance auditing (Huang and Hooker 2026):

Approach	What a "pass" means	Who must prove what	Risk in large samples	Risk in small samples
Significance testing	Failed to detect a violation	Regulator must find a violation	Flags trivially small gaps	Misses genuine gaps
TOST equivalence testing	Positive evidence gap is within tolerance	Insurer must demonstrate compliance	Narrow CIs help confirm fairness	Wide CIs yield "insufficient information"

Under TOST, a pass requires the confidence interval for the protected-group gap to lie fully inside pre-specified tolerance bands. In practice, the audit checks both:

1. **Ratio band** based on the adverse-impact tolerance (for example 0.80).
2. **Level-gap band** based on an allowed dollar or percentage gap (for example 5% of mean premium).

This produces three explicit outcomes — **pass**, **fail**, and **insufficient information** — with pre-specified escalation pathways for each. Insufficient information is not a pass; it signals that more data or further investigation is required.

6.4.4 4.4.4 Proxy discrimination testing

PD tests whether a rating variable acts as a statistical substitute for the protected attribute. It compares that variable's effect in a model without the protected attribute and a model with it.

A flag is raised when two conditions hold simultaneously: (a) the coefficient shift is statistically significant using the corrected cross-covariance standard error, and (b) the relative shift exceeds a minimum substantive threshold (for example 10%). This two-part rule separates statistical significance from practical materiality.

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7. 5. Case Studies

Three case studies provide technical implementation evidence. They are reproducible illustrations, not universal regulatory templates. Data, tolerance bands, and admissible factors remain jurisdiction- and product-specific.

7.1 5.1 Case Study 1: Fair Cost-Model Design

Data and setting. The case study uses the `pg15training` French motor insurance dataset (100,000 TPL policies from 2009–2010) from the CASdatasets R package (Dutang, Charpentier, et al. 2020), originally released for the 2015 French Institute of Actuaries pricing competition. The protected attribute is gender. Non-protected rating variables are: Age, Bonus, vehicle group (GroupOne), population density (Density), and vehicle value (Value). An insurance score — constructed from vehicle type, category, occupation, region, and age — is treated as the single non-legitimate variable, following the baseline scenario of Xin and Huang (2024).

The response variable is pure premium (expected claim cost), computed as claim frequency multiplied by claim severity. Frequency is modeled with Poisson GLM and severity with Gamma GLM; the XGBoost implementations use matching deviance losses.

Models. Five model designs from the taxonomy in Table 3 are implemented using both GLM and XGBoost: M0 (full, including gender), MU (gender excluded), MDP (all predictors debiased), MCDP (only insurance score debiased), and MC (full model with CPV averaging). Pre-processing uses the disparate impact remover of Feldman et al. (2015). Portfolio-level adjustments align total predicted premium for MDP, MCDP, MC, and all XGBoost models with the MU baseline, so comparisons focus on redistribution rather than overall portfolio level.

Evaluation metrics. Group fairness is measured by the disparate impact ratio (average predicted outcome in one group divided by the average in another). A value between 0.80 and 1.25 approximates the insurance four-fifths rule. Predictive accuracy is assessed by the normalized Gini index (ranking performance) and RMSE (overall error magnitude).

Illustrative findings. The fairness–accuracy analysis illustrates the expected trade-off: models that achieve better disparate impact ratios (closer to 1.0) tend to sacrifice some predictive accuracy, as measured by the Gini index or RMSE. The magnitude of the trade-off varies by model design and by whether GLM or XGBoost is used. MDP produces the sharpest premium redistribution across age and gender groups compared to the MU benchmark, while MCDP typically remains closer to MU because legitimate variables retain their predictive contribution. MC achieves proxy control beyond MU without explicit debiasing of the training data. The double-lift comparison of MU and MDP shows that policyholders where the two models disagree most (the tails of the premium ratio distribution) exhibit claim experience that tracks neither model perfectly — illustrating the adverse selection concern that accompanies cross-group premium redistribution.

7.2 5.2 Case Study 2: Welfare and Pricing-Rule Trade-Offs

Data and setting. Case Study 2 uses the same French motor insurance data as Case Study 1. Cost modeling (unconstrained XGBoost frequency-severity model, labeled C0) is estimated on the full 100,000-policy dataset. Welfare analysis uses a 2,000-policyholder subsample because the price optimization is computationally intensive at full scale. Gender is the protected attribute.

Framework. The welfare analysis follows Huang and Shimao (2026). Consumer demand is modeled with a discrete-choice setup: each policyholder chooses between the insurer's product and an uninsured outside option. Purchase probability is estimated using a logit demand model. Ex-ante product value is approximated with a mean-variance utility representation and parameters for risk aversion and inertia calibrated from prior literature (Cohen and Einav 2007; Jin, Stoline, et al. 2021).

Price optimization is implemented in Python using TensorFlow and the Adam optimizer (Kingma and Ba 2014), with gradient-based updating over up to 3,000 iterations per pricing rule. Fairness constraints for PDP and PAF are incorporated as penalty terms with weight 10^3.

Pricing rules evaluated. The five rules from Table 4 are applied with C0 cost inputs: C0-P0, C0-PA, C0-POB, C0-PDP, C0-PAF.

Illustrative findings. The baseline unconstrained pricing (P0) reveals an important asymmetry: female policyholders have lower estimated costs on average (and thus lower premiums), but higher markups ($p_i/\hat{c}_i - 1$) because the insurer extracts relatively more profit from this group. This illustrates that premium levels and markup levels can point in opposite directions — a lower price does not imply a lower relative burden.

Comparing regulations: PDP moves closest to premium equality between genders (by design) but increases female markups. PA reduces markup disparities while imposing larger profit losses than PDP. POB's welfare effects depend on voluntary vs. mandatory participation assumptions. PAF targets equity within risk strata, producing different redistribution patterns from PDP at the same overall premium level. Welfare change comparisons relative to P0 show that some regulations designed to help female policyholders can reduce their welfare through demand effects, and that no single rule simultaneously improves welfare and reduces markup disparity for the protected group.

7.3 5.3 Case Study 3: Deployment-Stage Fairness Testing

Data and setting. Case Study 3 uses the ProPublica Illinois auto insurance dataset (Larson et al. 2017), containing quoted premiums from multiple insurers across Illinois zip codes. The audit is conducted at the company level. The protected attribute is a majority-minority zip-code indicator ($A_z = 1$ for majority-minority). Legitimate rating factors are log state risk and a Chicago indicator. The audit follows the Plan → Audit → Decide → Improve protocol.

Audit parameters (pre-specified). Significance level $\alpha = 0.05$ (90% confidence intervals for TOST); minimum 50 zip codes per company; CDP dollar tolerance $\delta_{\%} = 5\%$ of mean premium (following Colorado draft guidance); ratio tolerance $\tau = 0.80$ (standard adverse-impact ratio); PD substantive threshold $\rho_{\text{min}} = 10\%$ relative coefficient shift.

CDP audit. For each company k, the audit regression is:

$$\log(P_{kz}) = \mu_{0k} + \beta_{A,k}A_z + \gamma_{1k}\log(\text{risk}_z) + \gamma_{2k}\text{Chicago}_z + r_{kz}$$

Both classical OLS and HC3 standard errors are computed. The HC3/classical SE ratio varies substantially across companies: ratios below 0.5 (classical overestimates uncertainty) and above 1.5 (classical underestimates uncertainty) are both observed. The direction is not predictable in advance and depends on the residual structure of each company's pricing function relative to the audit model. This confirms that corrected inference is not a refinement but a requirement.

TOST verdicts (pass / fail / insufficient information) are computed for each company using the HC3 confidence interval for $\beta_{A,k}$. Both the ratio band (derived from $\tau = 0.80$) and the level-gap band (5% of mean premium) must be satisfied for a pass. The distribution of verdicts across companies illustrates the three-outcome framework: some companies produce narrow enough confidence intervals to confirm compliance; others fail; and a third group has confidence intervals too wide to settle the question, triggering the insufficient-information verdict.

PD audit. The proxy discrimination test is applied to log risk as the suspected proxy variable. The audit compares the variable's coefficient with and without the protected attribute in the model, using the corrected cross-covariance standard error. Ignoring cross-covariance (treating the two estimates as independent) makes the PD test too conservative in this setting. Companies are flagged when the corrected two-part rule is met: statistical significance plus at least 10% relative shift.

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8. 6. Contributions

This preprint and the companion playbook contribute:

1. **An integrated governance workflow** — a four-step, sequentially linked framework from criterion definition through model design, welfare analysis, and audit, designed for use across compliance, actuarial, data science, and governance functions.
2. **A formal model taxonomy** — explicit mapping from fairness criteria (FTU, DP, CDP,

CPV) to model interventions (MU, MDP, MCDP, MC) at the cost-modeling stage, with mathematical definitions of each intervention and the criteria it targets.

3. **Welfare analysis as a required step** — the framework positions welfare evaluation (Step 3) as non-optional, not as an advanced supplement, because cost-model fairness does not mechanically produce premium fairness or consumer welfare improvements. The pricing rule comparison (P0, PA, POB, PDP, PAF) provides a concrete taxonomy for policy analysis.
4. **A statistically rigorous audit protocol** — pre-committed design, HC3-corrected inference for CDP, cross-covariance-corrected inference for PD, and equivalence testing (TOST) with a three-outcome classification (pass / fail / insufficient information). The protocol is grounded in the statistical properties of deterministic algorithmic outputs.
5. **Open implementation** — reproducible case studies in R (GLM and XGBoost for fair cost modeling) and Python (demand modeling, price optimization, fairness testing), available at github.com/feihuangFH/fair-pricing-playbook, with interactive guidance at fair.feihuang.org.

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9. 7. Limitations and Scope

Framework scope. The playbook is designed for operational clarity, not legal determination. It does not prescribe one globally valid fairness criterion, one approved model pipeline, or one set of acceptable rating factors. Legal burden, tolerated disparity, and admissible predictors vary across markets and product lines. The framework should not be treated as a compliance safe harbor in any specific jurisdiction.

Criterion selection. The case studies illustrate CDP and PD because these map to current regulatory proposals in Colorado and New York. Many other criteria — separation, sufficiency, individual fairness — may be appropriate in other jurisdictions or product contexts. The criterion that is legally binding must be determined by legal and regulatory counsel, not by the playbook.

Welfare analysis assumptions. The demand model in Case Study 2 relies on parameters calibrated from published studies (Cohen and Einav (2007); Jin, Stoline, et al. (2021)) and may not reflect the demand structure of other insurance markets, particularly markets with mandatory participation, different competitive structures, or different product types. The 2,000-policyholder subsample used for welfare analysis is a computational limitation.

Audit applicability. The HC3 correction and cross-covariance adjustment in Step 4 are designed for deterministic algorithmic outputs. For stochastic pricing systems (e.g., systems that incorporate random draws or exploration), different variance corrections may be needed. The majority-minority zip-code indicator used in Case Study 3 is a geographic proxy for race, not a direct protected-class observation; proxy measurement issues affect the interpretation of audit results as discussed in Xin, Hooker, and Huang (2026).

Partial dependence plots and feature importance. Some filing and governance workflows use variable-level interpretability tools (partial dependence plots, variable importance) as a fairness screen. These are distinct from CDP and PD: a variable-level plot that shows little marginal sensitivity to a protected attribute does not constitute a group-level fairness audit. The playbook treats interpretability tools as optional documentation supplements, not substitutes for outcome-based audit testing.

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10. 8. Conclusion

Fair algorithmic pricing requires more than model-level constraints. It requires coherent choices across fairness definition, model design, market impact assessment, and audit methodology. The Fair Pricing Playbook provides a practical, citable framework for that full pipeline. By linking criterion selection to model intervention (Step 2), welfare analysis (Step 3), and pre-committed audit protocols (Step 4), it offers a workflow that is defensible in governance review and actionable in practice. The companion case studies demonstrate that each step reveals something the prior steps cannot: cost-model fairness does not guarantee premium fairness, premium fairness does not guarantee consumer welfare improvements, and audit conclusions depend critically on whether inference is correctly specified for deterministic pricing outputs.

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12. Data Availability and Reproducibility

Case Study 1 uses the `pg15training` dataset from the CASdatasets R package (Dutang, Charpentier, et al. 2020), which is publicly available on CRAN. Case Study 3 uses the ProPublica Illinois auto insurance dataset, available at http://propublica.s3.amazonaws.com/projects/carinsurance/car-insurance-public.zip. Case Study 2 uses pre-computed optimization outputs derived from the French motor insurance data; the optimization code (Python/TensorFlow) is available in the companion repository.

Implementation code for all three case studies is available at github.com/feihuangFH/fair-pricing-playbook. The full interactive playbook is at fair.feihuang.org.

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13. 9. Access, Versioning, and Citation

- **Playbook site:** fair.feihuang.org
- **Source repository:** github.com/feihuangFH/fair-pricing-playbook
- **License:** CC BY 4.0
- **Version:** 1.0 (June 2026)

13.1 Suggested citation (APA)

Huang, F. (2026). *The Fair Pricing Playbook: A practical framework for developing, evaluating, and auditing fair algorithmic pricing systems* (Preprint, v1.0). [to be updated with DOI upon deposit]. fair.feihuang.org

13.2 BibTeX

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@misc{huang2026fairpricingplaybook,
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               developing, evaluating, and auditing fair algorithmic
               pricing systems},
  author    = {Huang, Fei},
  year      = {2026},
  note      = {Preprint, v1.0},
  doi       = {[to be assigned upon deposit]},
  url       = {https://fair.feihuang.org/preprint.html}
}
```

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